

5.1 Energy: Conservation, Work, Power & Efficiency

Question Paper

Course	CIE A Level Physics
Section	5. Work, Energy & Power
Topic	5.1 Energy: Conservation, Work, Power & Efficiency
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

Which of the following is a statement of the principle of conservation of energy?

- A. energy is the product of force and distance
- B. energy cannot be created, destroyed or transformed
- C. the total energy of an isolated system is constant
- D. the power transmitted by a system is proportional to the rate of transfer of energy

[1 mark]

Question 2

Which expression **defines** power?

- A. force $\times$ distance moved in the direction of the force
- B. force $\times$ velocity
- C. work done $\div$ time taken
- D. work done $\times$ time taken

[1 mark]

Question 3

The total energy input E_{in} in a process is partly transferred to useful energy output U , and partly to energy that is wasted W

What is the efficiency of the process?

- A. $\frac{U}{W} \times 100\%$
- B. $\frac{W}{E_{in}} \times 100\%$
- C. $\frac{U}{E_{in}} \times 100\%$
- D. $\frac{U + W}{E_{in}} \times 100\%$

[1 mark]

Question 4

A boat moving at constant speed v through still water experiences a total frictional drag F

What is the power developed by the boat?

A. $\frac{1}{2}Fv$

B. Fv

C. $\frac{1}{2}Fv^2$

D. Fv^2

[1 mark]

Question 5

What is the average power output of a laser that can deliver 0.20 J of energy in 10 ns?

A. 2 nW

B. 20 mW

C. 200 kW

D. 20 MW

[1 mark]

Question 6

A transformer has the following input and output

	Potential difference / V	Current / A
Input	11 000	28
Output	240	1200

What is the efficiency of the transformer?

- A. 0.94 %
- B. 1.0 %
- C. 11 %
- D. 94 %

[1 mark]

Question 7

Which operation involves the greatest mean power?

- A. a car moving against a resistive force of 0.4 kN at a constant speed of 20 m s^{-1}
- B. a crane lifting a weight of 3 kN at a speed of 2 m s^{-1}
- C. a crane lifting a weight of 5 kN at a speed of 1 m s^{-1}
- D. a weight being pulled across a horizontal surface at a speed of 6 m s^{-1} against a frictional force of 1.5 kN

[1 mark]

Question 8

An electric motor is required to produce 120 W of mechanical output power. The efficiency of the motor is 80%

Which row is correct?

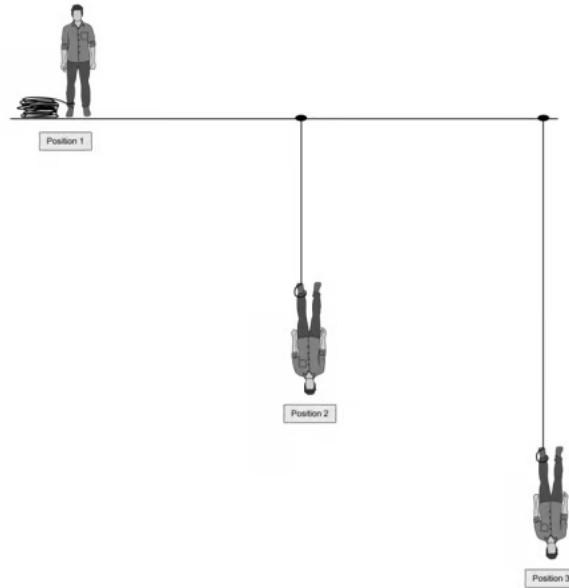
	Electrical power input to motor / W	Waste heat output from motor / W
A	120	24
B	120	96
C	150	30
D	150	120

A.

[1 mark]

Question 9

When bungee jumping, a student starts with maximum gravitational potential energy (position 1), then falls freely until the rope fully unwinds (position 2), after which the rope starts to stretch until the lowest point of the jump is reached (position 3)



What are the kinetic and elastic potential energies at position 3?

	Kinetic energy	Elastic potential energy
A	maximum	maximum
B	maximum	minimum
C	minimum	maximum
D	minimum	minimum

[1 mark]

Question 10

The kinetic energy of a moving vehicle with mass 1000 kg is 4.5×10^5 J. It brakes with a total constant braking force of 6000 N

What will be its stopping distance?

- A. 37 m
- B. 75 m
- C. 150 m
- D. 300 m

[1 mark]



Model Answers: Easy

1

The correct answer is **C** because:

- The Principle of Conservation of Energy states that energy cannot be created or destroyed, it can only be transferred from one form to another
- In other words, the total energy of an **isolated** system remains constant
- An isolated system is defined as a system that is closed off from its surroundings and this means that a system always has the same amount of energy, unless it's added from an external source
- Identifying the system is the most difficult part of conservation questions. Let's take a look at an example of a ball tossed up in the air:
- *During the throw:*
 - Chemical energy from your muscles → kinetic energy of the ball
- *As the ball reaches its maximum height:*
 - Kinetic energy of the ball → gravitational potential energy of the ball
- *As the ball falls down:*
 - Gravitational potential energy of the ball → kinetic energy of the ball
- The isolated system in this scenario is simply the ball, the muscles in the person's arm, and the Earth's gravitational field. It's not necessary to consider the chemical energy from the food the person ate or the gravitational fields of other bodies such as the sun or moon, for example

A & D are incorrect because these describe equations to work out work done and power, but they don't describe conservation of energy

B is incorrect because this is close to the classic definition 'energy cannot be created or destroyed; it can only be transferred from one form to another' but if you look closely you will see this definition states that energy cannot be transformed which is incorrect

2

The correct answer is **C** because:

- Power is defined as the rate of transfer of energy or work done
- In equation form this can be written as

$$\circ \text{ Power, } P = \frac{\text{energy}}{\text{time}} \text{ or } P = \frac{\text{work done}}{\text{time}}$$

A is incorrect because this equation describes how to calculate work done

B is incorrect because this is a correct formula for power but it derived from

$$\text{Power} = \frac{\text{work done}}{\text{time}}$$

D is incorrect because this is not a correct formula, it doesn't have a physical equivalent

3

The correct answer is **C** because:

- The efficiency of a system is the ratio of the energy used in the process to the total energy input in the system
- Efficiency = $\frac{\text{useful energy output}}{\text{total energy input}}$
- Useful energy output = U
- Total energy input = E_{in}
 - % Efficiency = $\frac{U}{E_{in}} \times 100\%$

4

The correct answer is **B** because:

- Power is defined as the rate of transfer of energy or work done
- In equation form this can be written as
 - Power, $P = \frac{\text{energy}}{\text{time}}$ or $P = \frac{\text{work done}}{\text{time}}$
- Work done = Force $\times$ Distance = $F \times d$
 - $P = \frac{\text{work done}}{\text{time}} = \frac{F \times d}{t} = Fv$
 - Where speed, $v = \frac{d}{t}$

- Therefore, $P = Fv$

5

The correct answer is **D** because:

- Power is defined as the rate of transfer of energy or work done
- This can be written as $P = \frac{\text{energy}}{\text{time}}$
- Time = 10 ns = 10×10^{-9} s
 - $P = \frac{0.20}{10 \times 10^{-9}} = 2 \times 10^7 \text{ W}$
- 1 MW = 1×10^6 W
 - $P = 2 \times 10^7 \text{ W} = 20 \text{ MW}$
- Therefore, the average power output of the laser 20 MW

6

The correct answer is **D** because:

- The efficiency of a transformer is the ratio of the power output of the process to the total power input to the system
- Efficiency = $\frac{\text{useful power output}}{\text{total power input}}$
- Power can be calculated using : $P = VI$
- Useful power output = $240 \times 1200 = 288000$
- Total power input = $11\,000 \times 28 = 308000$
 - % Efficiency = $\frac{\text{useful power output}}{\text{total power input}} \times 100\%$
 - % Efficiency = $\frac{288000}{308000} \times 100\% = 93.5\%$
- Therefore, the efficiency of the transformer is 94%

7

The correct answer is **D** because:

- Average, or mean, power is defined as the rate of transfer of energy or work done
- In equation form this can be written as

- Power, $P = \frac{\text{energy}}{\text{time}}$ or $P = \frac{\text{work done}}{\text{time}}$
- Work done = Force $\times$ Distance = $F \times d$
 - $P = \frac{\text{work done}}{\text{time}} = \frac{F \times d}{t} = Fv$
- In scenario D:
 - Force, $F = 1.5 \text{ kN}$
 - Speed, $v = 6 \text{ m s}^{-1}$
- $P = Fv = 1.5 \times 6 = 9 \text{ kN}$
- This operation gives the largest mean power, see the incorrect table for the other calculations

A is incorrect because $P = Fv = 0.4 \times 20 = 8 \text{ kN}$

B is incorrect because $P = Fv = 3 \times 2 = 6 \text{ kN}$

C is incorrect because $P = Fv = 5 \times 1 = 5 \text{ kN}$

8

The correct answer is **C** because:

- Efficiency = $\frac{\text{useful mechanical output power}}{\text{total electrical power}} = 80\%$
- Useful mechanical output power = 120 W
- Input power = $\frac{\text{useful mechanical output power}}{\text{efficiency}} = \frac{120}{0.8} = 150 \text{ W}$
- Waste heat power = input – output = 150 – 120 = 30 W

9

The correct answer is **C** because:

- From the conservation of energy, the sum of energies at position 2 or 3 should be equal to the initial amount of gravitational potential energy the student possesses at position 1

8

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9

The correct answer is **C** because:

- From the conservation of energy, the sum of energies at position 2 or 3 should be equal to the initial amount of gravitational potential energy the student possesses at position 1
- At position 2, the rope fully unwinds but it does not stretch. So, it does not have elastic potential energy. The kinetic energy is maximum at position 2
- At position 3, the rope is stretched. So, the elastic potential energy will reach a maximum
- At this position, the lowest point of the jump is reached, so, the student will not move any further, thus, the kinetic energy will be at a minimum

10

The correct answer is **B** because:

- Work done to stop vehicle = kinetic energy = $4.5 \times 10^5 \text{ J}$
- Work done = Force × Distance
- Distance = $\frac{\text{work done}}{\text{force}} = \frac{4.5 \times 10^5}{6000} = 75 \text{ m}$